19 Automated Reading of Flattened Alluvial Landscapes: A Comparative Application of Open LiDAR DTMs in Emilia-Romagna

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In intensely cultivated alluvial plains, mechanised deep ploughing and land levelling have largely erased archaeological micro-topography. In such settings, LiDAR rarely enables the direct detection of individual archaeological features, unlike in upland and/or forested environments. Nevertheless, subtle elevation differences preserved in Digital Terrain Models (DTMs) offer an invaluable archive for landscape-scale geoarchaeological analysis, revealing fluvial and beach ridges, as well as palaeochannels, which are often closely linked to settlement patterns.

This paper presents a reproducible workflow designed to exploit recently released open-access airborne LiDAR DTMs across the Emilia-Romagna region in Northern Italy. The proposed approach aims to move beyond subjective visual inspection by integrating multi-scale relief visualisation with unsupervised classification. Specifically, topographic variations were enhanced using the Multi-Scale Relief Model (MSRM), an approach particularly suited to revealing differences across flat landscapes at varying spatial extents. The resulting rasters were subsequently fed into a k-means clustering algorithm to achieve a largely automated segmentation of geomorphological features. A comparative application to the wider hinterlands of Ravenna and Piacenza demonstrates the potential and challenges of this pipeline for the automated mapping of subtle landforms across alluvial plains. By relying on open datasets and open-source algorithms, this study offers a workflow for the interpretation of landscape formation dynamics, consequently helping to shed light on how these processes shape both existing and future archaeological knowledge.